Modulation Strategies for PWM Full Bridge Converter

Introduction

In the realm of power electronics, the efficient conversion of electrical energy is paramount, especially in applications like charging infrastructure where energy loss directly impacts operational costs and environmental footprint. Full-bridge DC-DC converters are workhorses in these systems, but their performance is often limited by switching losses, particularly at high switching frequencies. This lecture delves into various modulation strategies for Pulse Width Modulation (PWM) full-bridge converters, focusing on how these techniques can be employed to achieve "soft switching."

Soft switching is a critical concept that minimizes energy dissipation during the turn-on and turn-off transitions of power semiconductor devices. By carefully controlling the timing of these transitions, we can significantly reduce switching losses, thereby improving the overall efficiency and reliability of the converter. Understanding these modulation strategies is essential for designing high-performance power converters that meet the stringent demands of modern charging systems and other power conversion applications.

Core Concepts

Term	Definition	Significance
`Soft Switching`	A technique where voltage or current across a switch is zero during its turn-on or turn-off transition, minimizing switching losses.	Crucial for improving converter efficiency, especially at high switching frequencies, by reducing energy dissipation in power semiconductors.
`Hard Switching`	A switching event where a device turns on or off while both voltage and current are non-zero, leading to significant power losses.	Represents the conventional switching method, which is less efficient and generates more electromagnetic interference (EMI) compared to soft switching.
`Dead Time`		

Term	Definition	Significance
	A short, intentional delay between the turn-off of one switch and the turn-on of its complementary switch in a half-bridge leg.	Essential for preventing shoot-through (short circuit) and enabling soft switching by allowing device output capacitances to charge/discharge.
`Leading Leg`	One of the two half-bridge legs in a full-bridge converter, typically where the turn-on times of its switches (S1 and S2) are adjusted in the forward direction.	Provides a mechanism to introduce phase shift or adjust pulse width for modulation, often facilitating soft switching for its own devices.
`Lagging Leg`	The other half-bridge leg in a full-bridge converter, where the turn-on times of its switches (S3 and S4) are adjusted in the backward direction.	Complementary to the leading leg, its adjustments also contribute to overall modulation and soft switching, particularly for its own devices.
`Duty Ratio (D)`	The fraction of a switching period during which a switch is turned on, determining the average output voltage of the converter.	Directly controls the output voltage of the converter; modulation strategies aim to maintain this output voltage while achieving soft switching.
`Diagonal Switches`	Pairs of switches in a full-bridge converter that are typically turned on or off simultaneously (e.g., S1 and S4, or S2 and S3).	Their synchronous or asynchronous turn-off behavior is a key differentiator between various modulation strategies, impacting soft switching characteristics.
`VAB Voltage`	The voltage across the primary winding of the isolation transformer in a full-bridge converter.	Its waveform and average value determine the power transfer and output voltage; modulation strategies aim to keep its volt-second balance unchanged.

Detailed Analysis

1. Soft Switching Fundamentals and Full-Bridge Overview

The concept of soft switching is foundational to understanding advanced PWM strategies. In a half-bridge converter, for instance, soft switching can be achieved for one of the switches during turn-on. When a switch (e.g., S1) turns off, the current (I0) flowing out of the half-bridge can charge and discharge the output capacitances (C1, C2) of the switches during a carefully introduced "dead time." This process ensures that the voltage across the complementary switch (S2) drops to zero before it turns on, leading to Zero-Voltage Switching (ZVS) turn-on. The necessary condition for this is that the energy stored in the inductor ($LI0^2$) must be greater than the energy stored in the device output capacitances ($C1Vin^2 + C2Vin^2$). Additionally, sufficient dead time must be provided for the capacitors to fully charge and discharge. However, the other switch (S1) typically experiences hard switching during its turn-on, incurring finite losses and high dV/dt.

Extending this to a full-bridge converter, we have four switches (S1, S2, S3, S4) exciting the primary winding of an isolation transformer. The secondary side typically includes a rectifier, an inductor (L), and a capacitor (C) to produce the output voltage (V0). In a basic full-bridge operation, diagonal switches (S1 and S4, or S2 and S3) are turned on simultaneously. The output voltage is given by V0 = N * Vin * D, where N is the transformer turns ratio, Vin is the input voltage, and D is the duty ratio. While this formula holds for ideal components, real-world converters include parasitic elements like transformer leakage inductance and device output capacitances (C1, C2, C3, C4, and body diodes D1, D2, D3, D4). These parasitics, often ignored in basic analysis, become crucial for enabling soft switching.

2. The Role of Leading and Lagging Legs in Modulation

To implement advanced modulation strategies, the two half-bridge legs of the full-bridge converter are often designated as the "leading leg" and the "lagging leg." This distinction is based on how their switch turn-on/off times are adjusted. The primary goal of these adjustments is to achieve soft switching for all or most switches without altering the converter's output voltage. This is achieved by ensuring that the voltage across the primary winding (VAB) remains unchanged during the adjustment periods. If VAB is maintained, the volt-second balance across the output inductor L remains the same, thus preserving the output voltage V0 = N * Vin * D.

The adjustments involve modifying the turn-on period of the switches. For the leading leg (e.g., S1 and S2), adjustments typically involve increasing the turn-on time in the "forward direction" (i.e., extending the pulse width). For the lagging leg

(e.g., S3 and S4), adjustments involve increasing the turn-on time in the "backward direction" (i.e., delaying the falling edge of the pulse). During these adjustment periods, the VAB voltage across the transformer primary is intentionally kept at zero. This is because, even if one switch's timing is altered, the states of the other three switches are such that no voltage is applied across the primary, or the current freewheels through the rectifier diodes.

3. Leading Leg Modulation Possibilities

The leading leg, comprising switches S1 and S2, offers three distinct possibilities for adjusting its turn-on period:

- **Possibility 1 (Unchanged On-time):** The turn-on time for S1 and S2 remains the standard DTs/2, where D is the duty ratio and Ts is the switching period. No adjustment is made to the leading edge of the pulses.
- **Possibility 2 (Forward Adjustment, On-time < Ts/2):** The turn-on period of S1 and S2 is adjusted forward by a small duration, delta T. The new on-time becomes DTs/2 + delta T. However, this adjusted on-time is still less than Ts/2. The delta T is less than (1-D)Ts/2. During this delta T period, the VAB voltage remains zero because the other switches are off, preventing any voltage from being applied to the primary.
- **Possibility 3 (Forward Adjustment, On-time = Ts/2):** The turn-on period of S1 and S2 is adjusted forward such that the total on-time becomes equal to Ts/2. This means delta T is precisely equal to (1-D)Ts/2. Again, during the adjustment period, VAB remains zero, ensuring the output voltage is unaffected.

4. Lagging Leg Modulation Possibilities

Similarly, the lagging leg, comprising switches S3 and S4, also offers three possibilities, but these involve adjusting the turn-on time in the backward direction (delaying the turn-off):

- **Possibility A (Unchanged On-time):** The turn-on time for S3 and S4 remains the standard DTs/2. No adjustment is made to the falling edge of the pulses.
- **Possibility B (Backward Adjustment, On-time < Ts/2):** The turn-on period of S3 and S4 is adjusted backward by a small duration, delta T. The new on-time becomes DTs/2 + delta T. This adjusted on-time is still less than Ts/2, with delta T less than (1-D)Ts/2. During this delta T period, VAB remains zero.

- **Possibility C (Backward Adjustment, On-time = Ts/2):** The turn-on period of S3 and S4 is adjusted backward such that the total on-time becomes equal to Ts/2. This means delta T is precisely equal to (1-D)Ts/2. As with the leading leg, VAB remains zero during the adjustment, preserving the output voltage.

5. Nine Modulation Methods and Their Classification

By combining the three possibilities from the leading leg (1, 2, 3) with the three possibilities from the lagging leg (A, B, C), a total of nine distinct PWM modulation strategies can be derived for the full-bridge converter. Each method aims to achieve soft switching for different sets of switches or under different operating conditions, all while maintaining the desired output voltage V0 = N * Vin * D.

These nine methods can be broadly categorized into two types based on the turn-off behavior of the diagonal switches:

- **Type 1: Diagonal Switches Turn Off at the Same Time (Methods 1, 2, 3)**
- **Method 1 (1A):** No changes are made to either the leading or lagging leg's turn-on times. This is the basic PWM operation where S1 and S4 turn off simultaneously, and S2 and S3 turn off simultaneously.
- **Method 2 (2A):** The leading leg's turn-on time is adjusted forward (Possibility 2), while the lagging leg's turn-on time remains unchanged (Possibility A). Diagonal switches S1/S4 and S2/S3 still turn off at the same time.
- **Method 3 (3A):** The leading leg's turn-on time is adjusted forward to Ts/2 (Possibility 3), while the lagging leg's turn-on time remains unchanged (Possibility A). Diagonal switches S1/S4 and S2/S3 still turn off at the same time.
- **Type 2: Diagonal Switches Turn Off at Different Times (Methods 4, 5, 6, 7, 8, 9)**
- **Method 4 (1B):** The leading leg's turn-on time is unchanged (Possibility 1), while the lagging leg's turn-on time is adjusted backward (Possibility B). Here, the diagonal switches S1/S4 and S2/S3 turn off at different times.
- **Method 5 (2B):** Both the leading leg (Possibility 2) and the lagging leg (Possibility B) have their turn-on times adjusted. Diagonal switches turn off at different times.
- **Method 6 (3B):** The leading leg's turn-on time is adjusted to Ts/2 (Possibility 3), and the lagging leg's turn-on time is adjusted backward (Possibility B). Diagonal switches turn off at different times.

- **Method 7 (1C):** The leading leg's turn-on time is unchanged (Possibility 1), while the lagging leg's turn-on time is adjusted backward to Ts/2 (Possibility C). Diagonal switches turn off at different times.
- **Method 8 (2C):** The leading leg's turn-on time is adjusted forward (Possibility 2), and the lagging leg's turn-on time is adjusted backward to Ts/2 (Possibility C). Diagonal switches turn off at different times.
- **Method 9 (3C):** Both the leading leg (Possibility 3) and the lagging leg (Possibility C) have their turn-on times adjusted to Ts/2. Diagonal switches turn off at different times.

In all these nine methods, the critical aspect is that the VAB voltage across the transformer primary remains unchanged during the adjustment periods. This ensures that the volt-second balance across the output inductor is maintained, and consequently, the output voltage V0 remains constant at N * Vin * D. The primary benefit of these diverse strategies lies in their ability to facilitate soft switching for different switches within the converter, thereby reducing overall switching losses and improving efficiency, particularly crucial for high-frequency operation where passive component sizes can be minimized.

Key Takeaways

- [ ] Soft switching minimizes switching losses by ensuring zero voltage or current across a switch during transitions.
- [ ] Hard switching occurs when both voltage and current are non-zero during transitions, leading to higher losses.
- [ ] Dead time is crucial in half-bridges to prevent shoot-through and enable capacitor charge/discharge for soft switching.
- [ ] Full-bridge converters use four switches, an isolation transformer, and a rectifier to produce a DC output voltage.
- [ ] Parasitic elements like leakage inductance and device output capacitances are essential for soft switching.
- [ ] Modulation strategies adjust switch turn-on/off times in leading and lagging legs without changing the output voltage (V0 = N * Vin * D).
- [ ] The VAB voltage across the transformer primary must remain unchanged during modulation adjustments to maintain output voltage.
- [ ] Nine distinct modulation methods arise from combining three leading leg and three lagging leg adjustment possibilities.
- [ ] These methods are categorized by whether diagonal switches turn off at the same time (Methods 1-3) or at different times (Methods 4-9).

- [ ] The ultimate goal of these strategies is to achieve soft switching, reducing losses and improving converter efficiency, especially at high switching frequencies.

Conclusion

The detailed exploration of modulation strategies for PWM full-bridge converters reveals a sophisticated approach to optimizing power conversion. By understanding and applying these nine distinct methods, designers can precisely control the switching behavior of power devices, moving from lossy hard switching to highly efficient soft switching. This capability is not merely an academic exercise; it directly translates into tangible benefits for real-world applications, particularly in demanding fields like electric vehicle charging infrastructure, where efficiency, reliability, and thermal management are paramount.

The ability to maintain a constant output voltage while strategically adjusting switch timings for soft switching underscores the elegance of these techniques. As power electronics continue to evolve towards higher switching frequencies and greater power densities, the mastery of such modulation strategies will be indispensable. The next step in this learning journey will involve delving into the practical implementation of these strategies and analyzing how they specifically enable soft switching for the leading and lagging leg switches, further solidifying their role in the design of next-generation power converters.